Exercises from Section 1: Metric Spaces, Gamelin and Greene

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Open and Closed Sets

Exercise 0.1. (Exercise 1)
Let U, V, and W be subsets of some set. Recall that $U \setminus V$ consists of all points in U that do not belong to V.

a) Prove that $(U \cup V) \setminus W = (U \setminus W) \cup (V \setminus W)$.

b) Prove that $(U \cap V) \setminus W = (U \setminus W) \cap (V \setminus W)$.

c) Does $U \setminus (V \setminus W)$ coincide with $(U \setminus V) \setminus W$? Justify your answer by proof or counterexample.

d) Prove the two set-theoretic identities used in the proof of Theorem 1.9.

Proof. a) Let $x \in (U \cup V) \setminus W$. Then

$$\begin{aligned} & x \in U \cup V \quad \text{and} \quad x \notin W \\ \iff & x \in U \text{ and } x \notin W \text{ or } x \in V \text{ and } x \notin W \\ \iff & x \in U \setminus W \quad \text{or} \quad x \in V \setminus W \\ \iff & x \in (U \setminus W) \cup (V \setminus W) \end{aligned}$$

b) Let $x \in (U \cap V) \setminus W$. Then

$$\begin{aligned} & x \in U \cap V \quad \text{and} \quad x \notin W \\ \iff & x \in U \text{ and } x \in V \text{ and } x \notin W \\ \iff & x \in U \text{ and } x \notin W \text{ and } x \in V \text{ and } x \notin W \\ \iff & x \in U \setminus W \quad \text{and} \quad x \in V \setminus W \\ \iff & x \in (U \setminus W) \cap (V \setminus W) \end{aligned}$$

c) Let $x \in U \setminus (V \setminus W)$. Then

$$\begin{aligned} & x \in U \quad \text{and} \quad x \notin V \setminus W \\ \iff & x \in U \quad \text{and} \quad x \in W \end{aligned}$$

Now let $x \in (U \setminus V) \setminus W$. Then

$$
\begin{aligned}
&x \in U \setminus V \quad \text{and} \quad x \notin W \\
\iff &x \in U \text{ and } x \notin V \text{ and } x \notin W
\end{aligned}
$$

Thus, $U \setminus (V \setminus W)$ and $(U \setminus V) \setminus W$ do not coincide.

d) The statements we prove that are used in the proof of the in-text Theorem 1.9 are as follows:

$$
X \setminus \bigcap_{\alpha \in A} E_\alpha = \bigcup_{\alpha \in A} (X \setminus E_\alpha) \quad \text{and} \quad X \setminus \bigcup_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} (X \setminus E_\alpha)
$$

with X a metric space with E_α subsets of X. Thus, let $x \in X \setminus \bigcap_{\alpha \in A} E_\alpha$. Then

$$
\begin{aligned}
&x \in X \quad \text{and} \quad x \notin \bigcap_{\alpha \in A} E_\alpha \\
\iff &x \in X \quad \text{and} \quad \exists \alpha \in A \text{ s.t. } x \notin E_\alpha \\
\iff &\exists \alpha \in A \text{ s.t. } x \in X \setminus E_\alpha \\
\iff &x \in \bigcup_{\alpha \in A} (X \setminus E_\alpha).
\end{aligned}
$$

Now for the second claim. Let $x \in X \setminus \bigcup_{\alpha \in A} E_\alpha$. Then

$$
\begin{aligned}
&x \in X \quad \text{and} \quad x \notin \bigcup_{\alpha \in A} E_\alpha \\
\iff &x \in X \quad \text{and} \quad x \notin E_\alpha \ \forall \alpha \in A \\
\iff &x \in X \setminus E_\alpha \ \forall \alpha \in A \\
\iff &x \in \bigcap_{\alpha \in A} (X \setminus E_\alpha)
\end{aligned}
$$

□

Exercise 0.2. (Exercise 2)
Show that (1.7) defines a metric on X. Show that every subset of the resulting metric space is both open and closed.

Proof. The metric mentioned in (1.7) is the discrete metric defined as

$$
d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}
$$

for all $x, y \in X$. We immediately see that $d(x, y) \geq 0$ for all $x, y \in X$. Likewise, $d(x, y) = 0$ if and only if $x = y$, by definition. If $x = y$, then $d(x, y) = d(y, x) = 0$ and if $x \neq y$, then $d(x, y) = d(y, x) = 1$. Finally, the triangle holds, with the only somewhat tedious verification being that $1 \leq 0 + 0 = 0$ being impossible (obviously). This occurs if $x \neq z$ and $x = y = z$ which would imply that $z \neq z$ which is an absurdity. Hence, d defines a metric on X. Since every set in a discrete metric space is both open and closed, every subset of X must be both open and closed.

□

Exercise 0.3. (Exercise 3)

a) Show that if $a, b, c \in \mathbb{R}$ are such that for all $\lambda \in \mathbb{R}$, $a\lambda^2 + b\lambda + c \geq 0$, then $b^2 - 4ac \leq 0$. *Hint:* Find the minimal value of the polynomial in λ.

b) Show that for any $(x_1, \ldots, x_n), (y_1, \ldots, y_n) \in \mathbb{R}^n$,

$$\sum_{i=1}^{n} x_i y_i \leq \left(\sum_{i=1}^{n} x_i^2\right)^{\frac{1}{2}} \left(\sum_{i=1}^{n} y_i^2\right)^{\frac{1}{2}}.$$

Remark: This is a version of the important Cauchy-Schwarz inequality. (Perhaps the most common spelling error in mathematics is to replace "Schwarz" by "Schwartz") For the proof, apply part (a) to

$$\sum_{i=1}^{n} (x_i - \lambda y_i)^2.$$

c) Using the Cauchy-Schwarz inequality, show that the function d defined by (1.5) satisfies the triangle inequality.

Proof. a) Assume $a, b, c \in \mathbb{R}$ such that for all $\lambda \in \mathbb{R}$, $f(\lambda) = a\lambda^2 + b\lambda + c \geq 0$. Consider the derivative $f'(\lambda) = 2a\lambda + b$. Setting this equal to 0, to obtain the minimal value's λ component, we get $\lambda = \frac{-b}{2a}$. Thus,

$$\begin{aligned} f\left(\frac{-b}{2a}\right) &= a\left(\frac{-b}{2a}\right)^2 + b\left(\frac{-b}{2a}\right) + c \\ &= \frac{b^2}{4a} - \frac{b^2}{2a} + c \\ &= \frac{b^2 - 2b^2 + 4ac}{4a} = \frac{-b^2 + 4ac}{4a} \end{aligned}$$

Since $f(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$, we have

$$\begin{aligned} 0 &\leq \frac{-b^2 + 4ac}{4a} \\ 0 &\leq -b^2 + 4ac \\ 0 &\geq b^2 - 4ac \end{aligned}$$

as desired.

b) Let $(x_1, \ldots, x_n), (y_1, \ldots, y_n) \in \mathbb{R}^n$ and let $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} \sum_{i=1}^{n} (x_i - \lambda y_i)^2 &= \sum_{i=1}^{n} x_i^2 - 2\lambda x_i y_i + \lambda^2 y_i^2 \\ &= \sum_{i=1}^{n} x_i^2 - 2\lambda \sum_{i=1}^{n} x_i y_i + \lambda^2 \sum_{i=1}^{n} y_i^2. \end{aligned}$$

Observe that this is a nonnegative quadratic in λ. Applying part (a),

$$\begin{aligned} \left(-2\sum_{i=1}^{n} x_i y_i\right)^2 - 4\left(\sum_{i=1}^{n} y_i^2\right)\left(\sum_{i=1}^{n} x_i^2\right) \leq 0 \\ 4\left(\sum_{i=1}^{n} x_i y_i\right)^2 &\leq 4\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) \\ \sum_{i=1}^{n} x_i y_i &\leq \left(\sum_{i=1}^{n} x_i^2\right)^{1/2} \left(\sum_{i=1}^{n} y_i^2\right)^{1/2}. \end{aligned}$$

Hence, we have proved this version of the Cauchy-Schwarz inequality.

c) The metric (1.5) is given by

$$d(x,y) = \left(\sum_{i=1}^{n}(x_i - y_i)^2\right)^{1/2}, \quad x, y \in \mathbb{R}^n.$$

Thus, let $x, y, z \in \mathbb{R}^n$. Then

$$\begin{aligned}
d(x,z) &= \left(\sum_{i=1}^{n}(x_i - z_i)^2\right)^{1/2} \\
&= \left(\sum_{i=1}^{n}(x_i - y_i + y_i - z_i)^2\right)^{1/2} \\
&= \left(\sum_{i=1}^{n}((x_i - y_i) + (y_i - z_i))^2\right)^{1/2} \\
&= \left(\sum_{i=1}^{n}(x_i - y_i)^2 + 2(x_i - y_i)(y_i - z_i) + (y_i - z_i)^2\right)^{1/2}.
\end{aligned}$$

Squaring both sides,

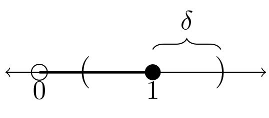
$$\begin{aligned}
d(x,z)^2 &= \sum_{i=1}^{n}(x_i - y_i)^2 + 2\sum_{i=1}^{n}(x_i - y_i)(y_i - z_i) + \sum_{i=1}^{n}(y_i - z_i)^2 \\
&\le \sum_{i=1}^{n}(x_i - y_i)^2 + 2\left(\sum_{i=1}^{n}(x_i - y_i)^2\right)^{1/2}\left(\sum_{i=1}^{n}(y_i - z_i)^2\right)^{1/2} + \sum_{i=1}^{n}(y_i - z_i)^2 \\
&= \left(\left(\sum_{i=1}^{n}(x_i - y_i)^2\right)^{1/2} + \left(\sum_{i=1}^{n}(y_i - z_i)^2\right)^{1/2}\right)^2 \\
&= d(x,y) + d(y,z). \quad \text{(taking square roots)}
\end{aligned}$$

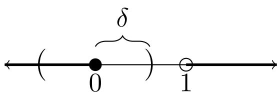
Note that the inequality is introduced due to the application of the Cauchy-Schwarz inequality of part (b). Therefore, (1.5) satisfies the triangle inequality.

□

Exercise 0.4. (Exercise 4)
Show that the semiopen interval $(0, 1]$ is neither open nor closed.

Proof. We use the analysis definition of openness. Hence, a set $U \subseteq \mathbb{R}$ is open if for all $x \in U$, there exists some $\delta > 0$ such that $(x - \delta, x + \delta) = V_\delta(x) \subseteq U$. Notice immediately that $V_\delta(1) \not\subseteq (0, 1]$. Conversely, the complement of $(0, 1]$ is $(-\infty, 0] \cup (1, \infty)$, which we see is not closed since $V_\delta(0) \not\subseteq (0, 1]^c$. Hence, $(0, 1]$ is neither open nor closed. (See visuals below for both cases.





□

Exercise 0.5. (Exercise 6)
Prove that the interior of a subset Y of X coincides with the union of all open subsets of X that are contained in Y. (Thus the interior of Y is the largest open set contained in Y.)

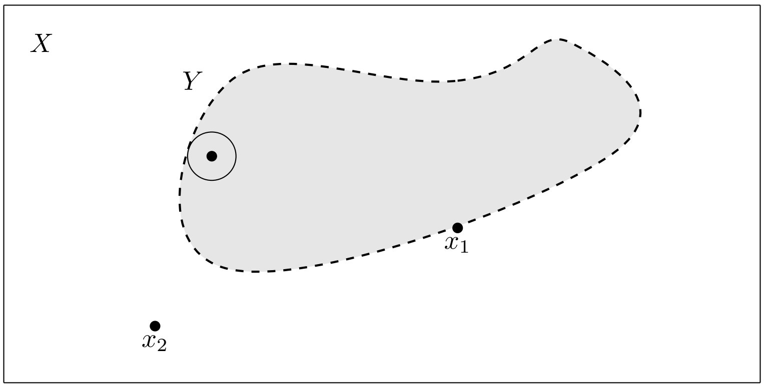
Proof. We want to show that $\text{Int}(Y) = \bigcup_{\alpha \in A} U_\alpha$ such that $U_\alpha \subseteq X$ is open and $U_\alpha \subseteq Y$. Then

$$\text{Int}(Y) = \{x \in X : \exists r > 0 \text{ s.t. } B(x;r) \subseteq Y\}$$

Let $x \in \text{Int}(Y)$. Then, there exist some $r > 0$ such that $B(x;r) \subseteq Y$. Since open balls in a metric space are open subsets of X, we associate $B(x;r)$ with the open set $U_{\alpha_1} \subseteq X$. Note that since $B(x;r) \subseteq Y$, we have $U_{\alpha_1} \subseteq Y$. Hence, by induction on all of the elements of $\text{Int}(Y)$, we associate some open subset with the open ball centered around that element. The converse correspondence is also true (Theorem 1.1). Hence

$$\text{Int}(Y) = \bigcup_{\substack{U \subseteq X \\ \text{open} \\ U \subseteq Y}} U.$$

(See the diagram below)



$x_1 \notin \text{Int}(Y)$ since there is no $r > 0$ such that $B(x_1;r) \subseteq Y$.
$x_2 \notin \text{Int}(Y)$ since there is no $r > 0$ such that $B(x_2;r) \subseteq Y$
$x_3 \in \text{Int}(Y)$ since there is an $r > 0$, one could choose the minimum distance to the boundary of the curve, such that $B(x_3;r) \subseteq Y$

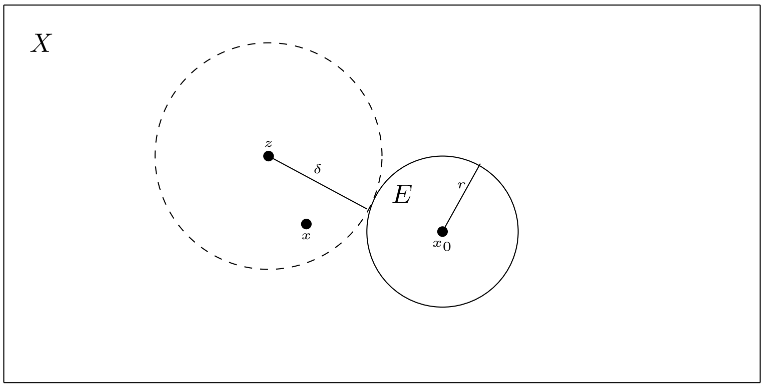
□

Exercise 0.6. (Exercise 8)
A set of the form $\{y \in X : d(x,y) \leq r\}$ is called a closed ball. Show that a closed ball is a closed set. Is the closed ball $\{y \in X : d(x,y) \leq r\}$ always the closure of the open ball $B(x;r)$? What if $X = \mathbb{R}^n$? Prove your answers.

Proof. Let E be the closed ball centered at $x_0 \in X$ with radius $r > 0$. Hence,
$E = \{x \in X : d(x,x_0) \leq r\}$. Let $z \in X \setminus E$. Set $\delta = d(z,x_0) - r > 0$. Suppose $x \in B(z,\delta)$. Then

$$d(x,x_0) \geq d(z,x_0) - d(z,x) > d(z,x_0) - \delta = r$$

Hence, $x \notin E \Rightarrow x \in X \setminus E$. Thus, $B(z,\delta) \subseteq X \setminus E$. Then there is an open ball for every $x \in X \setminus E$ and therefore $X \setminus E$ is open and thus E is closed.
NEED TO DO SECOND PART!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!



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0.1 Products of Metric Spaces

Exercise 0.7. (Exercise 1)
Prove that the functions on $X \times X$ given by (4.1),(4.2), and (4.3) are metrics.

Proof. Throughout this proof, let $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n), z = (z_1, \ldots, z_n) \in X = \prod_{i=1}^{n} X_i$ with each X_i having a metric d_i.
(4.1): Define $d \colon X \times X \to X$ by $d(x, y) = \left(\sum_{i=1}^{n} d_i(x_i, y_i)^2\right)^{1/2}$. Then since $d_i(x_i, y_i) \geq 0$ for all i, it follows that $d(x, y) \geq 0$. Let

$$\begin{aligned} x = y &\iff x_i = y_i \ \forall i \\ &\iff d_i(x_i, y_i) = 0 \ \forall i \\ &\iff d(x, y) = 0. \end{aligned}$$

Next,

$$\begin{aligned} d(x, y) &= \left(\sum_{i=1}^{n} d_i(x_i, y_i)^2\right)^{1/2} \\ &= \left(\sum_{i=1}^{n} d_i(y_i, x_i)^2\right)^{1/2} \\ &= d(y, x). \end{aligned}$$

Finally, we prove that the triangle inequality is satisfied. Set $a = (d_1(x_1, y_1), \ldots, d_n(x_n, y_n))$ and $b = (d_1(y_1, z_1), \ldots, d_n(y_n, z_n))$. Hence,

$$d(x, y) = \|a\|_2 \quad \text{and} \quad d(y, z) = \|b\|_2$$

with $\|\cdot\|_2$ the Euclidean norm. Since $d_i(x_i, z_i) \leq a_i + b_i$, we obtain

$$d(x, z) = \|(d_i(x_i, z_i))\|_2 \leq \|a + b\|_2.$$

Then

$$\begin{aligned} (\|a + b\|_2)^2 &= \sum_{i=1}^{n} a_i^2 + 2\sum_{i=1}^{n} a_i b_i + \sum_{i=1}^{n} b_i^2 \\ &= (\|a\|_2)^2 + 2\sum_{i=1}^{n} a_i b_i + (\|b\|_2)^2. \end{aligned}$$

By the Cauchy-Schwartz inequality, $\sum_{i=1}^{n} a_i b_i \leq \|a\|_2 \|b\|_2$. Thus,

$$(\|a+b\|_2)^2 \leq (\|a\|_2)^2 + 2\|a\|_2\|b\|_2 + (\|b\|_2)^2 = (\|a\|_2 + \|b\|_2)^2 .$$

Taking square roots,

$$\|a+b\|_2 \leq \|a\|_2 + \|b\|_2.$$

Hence,

$$\begin{aligned} d(x,z) &\leq \|a\|_2 + \|b\|_2 \\ &= d(x,y) + d(y,z). \end{aligned}$$

Therefore, (X,d) is a metric space.
(4.2): Define $d\colon X \times X \to X$ by $d(x,y) = \max_i(d_i(x_i,y_i))$. Since $d_i(x_i,y_i) \geq 0$ for all i, we must have $d(x,y) = \max_i(d_i(x_i,y_i)) \geq 0$. Let

$$\begin{aligned} x = y &\iff x_i = y_i \ \forall i \\ &\iff d_i(x_i,y_i) = 0 \ \forall i \\ &\iff d(x,y) = 0. \end{aligned}$$

Next, $d(x,y) = \max_i(d_i(x_i,y_i)) = \max_i(d_i(y_i,x_i)) = d(y,x)$. Finally, observe that

$$d_i(x_i,y_i) \leq \max_j(d_j(x_j,y_j)) \quad \text{and} \quad d_i(y_i,z_i) \leq \max_j(d_j(y_j,z_j))$$

for every $i = 1, \ldots, n$, so

$$d_i(x_i,y_i) + d_i(y_i,z_i) \leq \max_j(d_j(x_j,y_j)) + \max_j(d_j(y_j,z_j)).$$

This holds for all i, so in particular it holds for the index corresponding to the maximum. In other words,

$$\max_i(d_i(x_i,y_i) + d_i(y_i,z_i)) \leq \max_i(d_i(x_i,y_i)) + \max_i(d_i(y_i,z_i)).$$

Then since $d_i(x_i,z_i) \leq d_i(x_i,y_i) + d_i(y_i,z_i)$ for every i, by "applying" the maximum, we obtain

$$\begin{aligned} \max_i(d_i(x_i,z_i)) = d(x,z) &\leq \max_i(d_i(x_i,y_i) + d_i(y_i,z_i)) \\ &= \max_i(d_i(x_i,y_i)) + \max_i(d_i(y_i,z_i)) \\ &= d(x,y) + d(y,z). \end{aligned}$$

Therefore, (X,d) is a metric space.
(4.3): Define $d\colon X \times X \to X$ by $d(x,y) = \sum_{i=1}^{n} d_i(x_i,y_i)$. Thus, since $d_i(x_i,y_i) \geq 0$ for all i, it follows that $d(x,y) = \sum_{i=1}^{n} d_i(x_i,y_i) \geq 0$. Let

$$\begin{aligned} x = y &\iff x_i = y_i \ \forall i \\ &\iff d_i(x_i,y_i) = 0 \ \forall i \\ &\iff d(x,y) = 0. \end{aligned}$$

Next, $d(x,y) = \sum_{i=1}^{n} d_i(x_i,y_i) = \sum_{i=1}^{n} d_i(y_i,x_i) = d(y,x)$. Finally, $d_i(x_i,z_i) \leq d_i(x_i,y_i) + d_i(y_i,z_i)$ for every i. Hence,

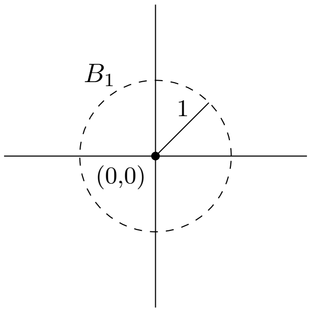
$$\begin{aligned}\sum_{i=1}^{n} d_i(x_i, z_i) = d(x, z) &\le \sum_{i=1}^{n} (d_i(x_i, y_i) + d_i(y_i, z_i)) \\ &= \sum_{i=1}^{n} d_i(x_i, y_i) + \sum_{i=1}^{n} d_i(y_i, z_i) \\ &= d(x, y) + d(y, z).\end{aligned}$$

Therefore, (X, d) is a metric space.

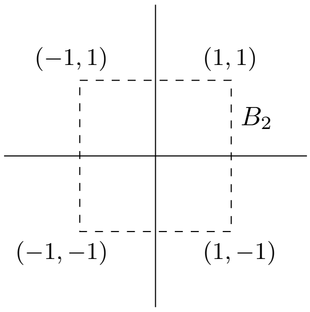
$\square$

Exercise 0.8. (Exercise 2)
For each of the metrics (4.1), (4.2), and (4.3) on the plane $\mathbb{R}^2$, sketch the open ball $B(0; 1)$. What inclusion relations hold between these balls?

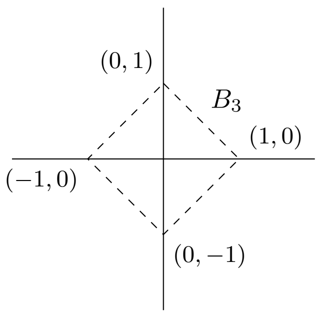
Proof. (4.1): Let $B_1 = B(0; 1) = \{y \in X : d(0, y) < 1\} = \{y \in X : (0^2 + y^2)^{1/2} < 1\} = \{y \in X : y < 1\}$. Hence,



(4.2): Let $B_2 = B(0; 1) = \{y \in X : d(0, y) < 1\} = \{y \in X : \max(|y_1|, |y_2|) < 1\} = \{y \in X : y_1, y_2 < 1\}$. Hence,



(4.3): Let $B_3 = B(0; 1) = \{y \in X : d(0, y) < 1\} = \{y \in X : |y_1| + |y_2| < 1\}$. Hence,



Therefore, we have the following inclusion relation

$$B_3 \subseteq B_1 \subseteq B_2.$$

$\square$

Normed Linear Spaces

Exercise 0.9. (Exercise 12)

a) Prove that if f and g are continuous real-valued functions on $[0,1]$, then

$$\int_0^1 f(s)g(s)ds \leq \left(\int_0^1 f(s)^2 ds\right)^{1/2} \left(\int_0^1 g(s)^2 ds\right)^{1/2}.$$

Remark: This is an integral version of the Cauchy-Schwarz inequality. For the proof, refer back to Exercise 1.3 and consider the integral

$$\int_0^1 [f(s) - \lambda g(s)]^2 ds,$$

which is nonnegative for all values of the real parameter λ.

b) Using (a), show the formula

$$\|f\|_2 = \left(\int_0^1 f(s)^2 ds\right)^{1/2}$$

defines a norm on the space of continuous real-values functions on $[0,1]$.

c) Prove that if f and g are continuous complex-values functions on $[0,1]$, then

$$\int_0^1 f(s)\overline{g(s)}ds \leq \left(\int_0^1 |f(s)|^2 ds\right)^{1/2} \left(\int_0^1 |g(s)|^2 ds\right)^{1/2}.$$

Remark: This is an integral form of the complex version of the Cauchy-Schwarz inequality.

d) Show that the formula

$$\|f\|_2 = \left(\int_0^1 |f(s)|^2 ds\right)^{1/2}$$

defines a norm on the space of continuous complex-valued functions on $[0,1]$.

Proof. a) Consider the integral

$$\int_0^1 [f(s) - \lambda g(s)]^2 ds \geq 0, \qquad \text{for all } \lambda \in \mathbb{R}.$$

Expanding the integrand, we obtain,

$$\begin{aligned}
\int_0^1 [f(s) - \lambda g(s)]^2 ds &= \int_0^1 f(s)^2 - 2\lambda f(s)g(s) + \lambda^2 g(s)^2 ds \\
&= \int_0^1 f(s)^2 ds - 2\lambda \int_0^1 f(s)g(s)ds + \lambda^2 \int_0^1 g(s)^2 ds \geq 0.
\end{aligned}$$

Observe that this is a quadratic in λ with real coefficients, and thus by Exercise 1.3,

$$\begin{aligned}\left(-2\int_0^1 f(s)g(s)ds\right)^2 - 4\left(\int_0^1 g(s)^2ds\right)\left(\int_0^1 f(s)^2ds\right) &\leq 0\\ 4\left(\int_0^1 f(s)g(s)ds\right)^2 &\leq 4\left(\int_0^1 g(s)^2ds\right)\left(\int_0^1 f(s)^2ds\right)\\ \int_0^1 f(s)g(s)ds &\leq \left(\int_0^1 g(s)^2\right)^{1/2}\left(\int_0^1 f(s)^2ds\right)^{1/2}.\end{aligned}$$

b) ??

c) ??

d) ??

□